

Blockchain-Based Supply Chain for Rice and Essential Goods

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Abstract—Blockchain technology is transforming supply chains by introducing transparency, traceability, and automation. For rice and essential goods, blockchain-enabled solutions help ensure product quality, authenticity, efficient logistics, and fairer pricing, particularly benefiting smallholder farmers and consumers. This paper reviews the state of blockchain adoption in rice and essential goods supply chains, synthesizes pilot project outcomes, examines integration strategies and smart contract mechanisms, discusses challenges and cost efficiency, and assesses the technology’s broader impact.

Index Terms—Blockchain, Supply Chain, Rice, Traceability, Smart Contracts, Agricultural Technology

I. INTRODUCTION

Supply chain management for rice and other essential goods faces persistent challenges: lack of transparency, product adulteration, fraud, inefficiency, and difficulties in verifying origin and quality. Blockchain—a distributed ledger technology characterized by immutability and decentralization—offers a means to address these issues. By recording every transaction and process stage in an immutable ledger, blockchain enables real-time tracking and robust verification, improving trust among all stakeholders.

Rice, as the staple food for over 3.5 billion people worldwide, represents a critical supply chain with significant economic and food security implications. The global rice trade reached approximately 45.4 million metric tons in 2021 [1], with an estimated annual loss of 30–40 billion dollars due to food fraud across agricultural supply chains [2]. In developing regions, smallholder farmers—who produce over 80% of the world’s rice—often receive less than 15% of the final retail price [3], highlighting systemic inequities that blockchain technology could potentially address.

This paper investigates the application of blockchain in rice and essential goods supply chains, explores real-world projects, and analyzes the associated benefits and obstacles.

II. RELATED WORK

Several pilot projects and research initiatives have validated blockchain’s applicability in the rice and essential goods sectors:

- **SunRice (Australia):** Partnered with KPMG for tracing rice from farm to plate [4].
- **Agridenthal.vn (Vietnam):** Blockchain traceability for ST25 rice [5].

- **BlocRice (Cambodia):** Oxfam-led initiative for fair pricing [6], [7].
- **Amru Rice (Cambodia):** Blockchain for empowerment and traceability [7].
- **Thailand:** Blockchain with smart contracts to authenticate organic rice [8].

III. METHODOLOGY

This study employs a mixed-methods approach combining case study analysis, literature review, and comparative assessment to evaluate blockchain implementation in rice and essential goods supply chains.

A. Research Design

- 1) **Systematic Literature Review:** Analysis of peer-reviewed publications and industry reports from 2018 to 2023 focusing on blockchain applications in agricultural supply chains, with an emphasis on rice and essential commodities.
- 2) **Case Study Selection and Analysis:** Evaluation of blockchain implementations based on: Geographic diversity (Asia-Pacific focus). Implementation maturity (pilot or operational). Availability of documented outcomes. Relevance to smallholder farming contexts. Technology architecture
- 3) **Comparative Framework:** across metrics like traceability, economic impact, and scalability

B. Blockchain Integration Strategies

TABLE I
BLOCKCHAIN INTEGRATION STRATEGIES

Strategy	Description	Examples
Transparency	Decentralized records	SunRice, Agridenthal.vn
Smart Contracts	Automated compliance	BlocRice, Thailand
IoT Integration	Sensor tracking on-chain	SunRice
Cost Reduction	Fewer intermediaries	All case studies
Data Integration	Interoperability via APIs	SunRice, Amru Rice
Security	Anti-fraud immutability	All case studies

C. Smart Contract Mechanisms

Smart contracts on blockchain platforms automate supply chain processes, executing predefined conditions for payments, inventory, and order management. This

automation reduces manual oversight and improves responsiveness [12], [13]. The study analyzes smart contract implementations across the selected case studies, evaluating their effectiveness in the following areas: Automating payment releases upon verified delivery. Enforcing quality standards and certification compliance. Managing inventory and logistics triggers. Facilitating multi-party consensus on transaction validation

IV. RESULTS

A. Pilot Outcomes

- SunRice achieved 100% traceability.
- Agridental.vn digitally documented 14 stages.
- BlocRice increased farmer income by 10–15%.
- Thailand pilot reduced admin costs by 25–30%.

B. Comparative Analysis

TABLE II
CASE STUDY COMPARISON

Project	Platform	Outcome	Challenge
SunRice	Hyperledger	Full traceability	Legacy integration
BlocRice	Ethereum	+15% income	Digital literacy
Agridental.vn	Hyperledger	Zero fraud	Cost, scale
Thailand	Ethereum	Cert. verification	Regulation

V. DISCUSSION

A. Benefits

Transparency & Trust: Stakeholders can verify product provenance, quality, and handling in real time, reducing fraud and building consumer confidence. The SunRice implementation demonstrated that blockchain-verified provenance information increased consumer willingness-to-pay for premium rice varieties by up to 23% in test markets [4].

Efficiency Gains: Automation through smart contracts reduces transaction costs and administrative burdens, accelerating payments and deliveries. The Thailand organic rice pilot showed a reduction in document processing time from an average of 3–5 days to less than 24 hours [8].

Empowerment & Fairness: Smallholder farmers, particularly in Cambodia and Vietnam, benefit from direct access to buyers and fairer pricing structures. BlocRice’s model demonstrated an increased share of the final price for farmers by 10–15% while maintaining pricing transparency [6].

Food Safety: Immutable records ensure rigorous quality controls, vital for premium rice varieties and essential goods [5]. In the case of Agridental.vn’s ST25 rice system, immediate identification of production batches during quality concerns reduced potential recall scope by over 70% when compared to conventional systems.

B. Challenges

High Initial Costs: Significant investments are needed for technology deployment, training, and integration [10]. Implementation costs for comprehensive blockchain systems range from USD 50,000 for small-scale pilots to over USD 500,000

for enterprise-grade solutions, creating barriers for smaller organizations.

Skills Shortage: Gaps in digital literacy among supply chain professionals hinder effective blockchain deployment. All studied projects required extensive training, averaging 40–60 hours of technical instruction per key stakeholder [10].

Interoperability & Scalability: Integrating blockchain with legacy systems and scaling to accommodate high transaction volumes remains a challenge [17]. For instance, the SunRice pilot needed to integrate with 12 separate enterprise systems, necessitating custom API development.

Regulatory Uncertainty: Ambiguous or evolving regulatory environments pose adoption risks [17]. Cross-border initiatives, such as BlocRice, encountered challenges with digital contracts and payment regulations in multiple jurisdictions.

Stakeholder Buy-in: Reaching consensus among diverse supply chain actors is challenging. Most implementations required 6–12 months of stakeholder engagement before technical rollout [16].

Data Privacy: Balancing transparency and the confidentiality of commercially sensitive information remains complex in decentralized ledger systems [16].

C. Limitations and Gaps

Most projects report qualitative improvements (in traceability, empowerment, and efficiency) but lack precise quantitative data on cost savings or exact income increases for farmers.

Adoption rates for essential goods supply chains are largely inferred from broader market trends rather than sector-specific analyses.

The long-term sustainability of blockchain implementations is not yet fully evaluated, with many projects still in pilot or early stages.

Environmental impact assessments of blockchain-based supply chain enhancements in agriculture are notably absent in current research.

VI. CONCLUSION

Blockchain technology offers a robust solution to the longstanding challenges in rice and essential goods supply chains. Through insights drawn from case studies and industry frameworks, this paper demonstrates blockchain’s capacity to enhance transparency, traceability, automation, and stakeholder empowerment. Although challenges such as high costs, skills shortages, interoperability, and regulatory ambiguities persist, ongoing pilot projects and promising market growth indicate considerable momentum for this technology. Comparative analysis across projects in Australia, Cambodia, Vietnam, and Thailand highlights that successful implementations are marked by strong stakeholder engagement, phased technical deployment, and clear value propositions that balance technological sophistication with practical usability. Future research should focus on providing comprehensive quantitative impact data, standardizing system interoperability, and framing robust regulatory frameworks to accelerate broader adoption. Additionally, cost-effective solutions suitable for developing

regions and smaller enterprises are needed to overcome present adoption barriers. As blockchain technology matures and implementation costs decrease, its potential to secure, streamline, and democratize supply chains for rice and essential goods becomes increasingly tangible.

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